

Instituto Tecnológico y de Estudios Superiores de Occidente

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Master of Data Science



Thesis Title

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MASTER OF DATA SCIENCE

A thesis presented by:
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Thesis Advisors:
Dr. Willie E. Coyote
Dr. Homer J. Simpson

Tlaquepaque, Jalisco, January, 2021

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Abstract

Here

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Resumen

En esta página debes escribir el abstract pero en español.

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1 Introduction

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1.1 Learning from Graph-Structured Data

Data in machine learning often comes in a few familiar forms. Most datasets are tabular: a table of rows and columns in which each row is an independent example described by the same fixed set of features. Other data is organized as images, that is, grids of pixels, or as sequences such as text and audio, whose elements follow one another in a specific order. What these forms share is regularity: each has a fixed, well-defined structure that repeats across every example.

Many real-world systems do not fit any of these shapes. A molecule is a set of atoms connected by chemical bonds, and neither the number of atoms nor the way they are connected is the same from one molecule to the next. A social network is a set of people connected by relationships, where each person may be linked to a different number of others. A citation network is a set of papers that reference one another, each citing a different set of others. These are not tables, images, or sequences; they are networks of entities and the relations between them (Figure 1.1). Data of this kind is naturally described by a *graph*: a set of entities, called *nodes*, together with a set of connections between them, called *edges*. Both nodes and edges can carry their own data, for example in a molecule, each node is an atom labeled with its element (carbon, hydrogen, oxygen, and so on) and each edge is a bond of a particular type (single, double, and so on); in a social network, the nodes are people and the edges are the relationships among them.

Rather than treating a domain as a collection of isolated data points, we describe it in terms of entities and the relationships among them, which means there is an underlying graph whose structure ties those

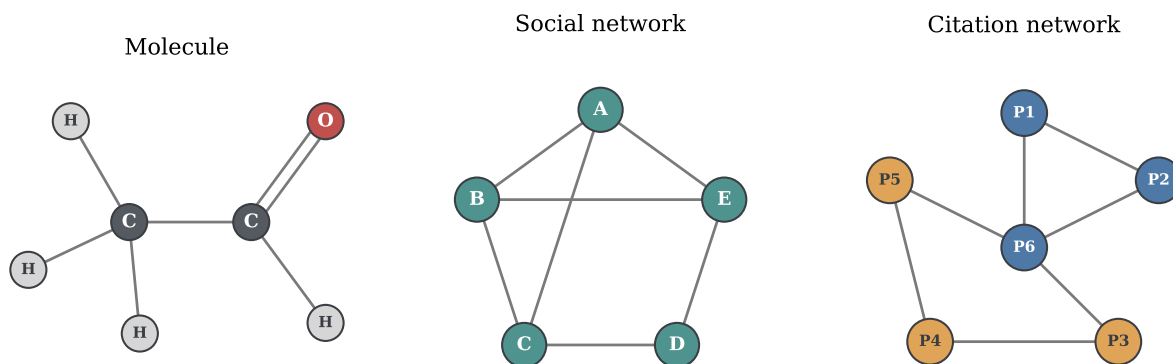


Figure 1.1: Three domains represented as graphs. A molecule, whose nodes are atoms labeled by their element and whose edges are bonds (single or double); a social network, whose nodes are people and whose edges are relationships; and a citation network, whose nodes are papers (colored by topic) and whose edges are citations between them. In each case the entities and their connections, not a table, grid, or sequence, define the data.

entities together. Seen this way, the graph is not merely one possible encoding of a molecule or a social network but a faithful representation: it records exactly which entities exist and how they relate, without forcing the data into a shape it does not have. A wide range of domains exhibit this kind of relational structure, and modeling them explicitly as graphs, rather than ignoring the connections, can lead to more accurate predictions. Data organized in this way is called *graph-structured data*.

Most deep learning models are tailored to the regular structures described above, particularly sequences and images laid out on a grid. Graphs are harder to work with. They have no fixed size, since different graphs contain different numbers of nodes. They have no fixed layout: a pixel in a grid has a clear sense of place, with neighbors above, below, and to either side, whereas a node in a graph has no such spatial arrangement and may be connected to any number of others. And they have no natural ordering, with no first or last node and no canonical point from which to read the rest.

The goal, then, is a model that takes a graph as input and produces predictions about it, such as a label for each node, whether connections exist between pairs of nodes, or a single label for the graph as a whole. This raises the central question: how can a predictive model – concretely, a neural network – be designed to operate directly on a graph, respecting its variable size and its lack of order instead of discarding them? Neural networks built for this purpose are called *Graph Neural Networks*, hereafter abbreviated as GNNs.

1.2 Objectives and Scope

The general objective of this work is to provide a self-contained reference on GNNs for homogeneous graphs under the *message-passing* framework: a single document that develops the theory that motivates

and defines these models, presents representative architectures, and evaluates them through reproducible implementations.

Specifically, the work aims to:

- establish the foundations that motivate GNNs: how graphs represent relational data, why standard neural networks cannot learn from it directly, and the inductive biases a model on graphs must respect;
- develop the message-passing framework as the unifying formulation from which GNN architectures are derived;
- present representative architectures for the three standard learning tasks on graphs: the Graph Convolutional Network (GCN) for node classification, the Graph Auto-Encoder (GAE) for link prediction, and the Graph Isomorphism Network (GIN) for graph classification;
- implement the three architectures on standard benchmark datasets (Cora, Amazon Computers, and NCI1) using PyTorch Geometric, with reproducible implementations, and evaluate each against a multilayer perceptron baseline that ignores the graph, examining the predictive benefit of explicitly modeling the relational structure of the data;
- examine the principal limitations of GNNs under this framework, including *oversmoothing* and the ceiling on their expressive power.

Its contribution is the reference document itself: a single, reproducible source that develops the message-passing framework from first principles, presents representative GNN architectures, and serves as a foundational resource for studying the field.

The graphs we consider are treated mainly as *undirected*, meaning their connections carry no direction, and *simple*, meaning that at most one connection joins any pair of nodes and that no node is connected to itself. This choice reflects simplicity rather than a limitation of the framework: message passing has enough design space to accommodate *directed* graphs, *self-loops* (a node connected to itself), and even *multigraphs* (more than one connection between the same pair of nodes).

The graphs considered in this work are also *homogeneous*: they contain a single kind of node and a single kind of connection. This restriction is deliberate. The homogeneous setting is where the message-passing framework is most directly understood, and it is the foundation on which more elaborate models are built, including *heterogeneous* graphs, which carry several kinds of nodes and connections and require machinery beyond the scope of this work. Establishing the homogeneous case is a prerequisite for these extensions, which we

leave as future research. The formal definitions of these graph types are given later in the document.

1.3 *Structure of the Document*

This introduction has set out the motivation, objectives, and scope of the work. The remainder of the document is organized as follows.

The first chapters establish the foundations. Chapter 2 surveys the tasks that graph learning addresses at the node, edge, and graph level, together with the practical applications that motivate the rest of the work. Chapter 3 establishes the formal definitions and notation for graphs used throughout. Chapter ?? reviews the multilayer perceptron and shows why standard neural networks are ill-suited to relational data. Chapter ?? develops the notion of inductive bias and the permutation invariance and equivariance that any graph layer must satisfy.

Chapter ?? builds on these foundations to derive the message-passing framework: the message, aggregation, and update functions, the receptive field that successive layers induce, the expressive power of the resulting models, and the architectural design space it enables.

Chapter ?? presents the three representative architectures within this framework: GCN for node classification, GAE for link prediction, and GIN for graph classification. Chapter ?? turns to practice, providing reproducible implementations of these architectures on the Cora, Amazon Computers, and NCI1 benchmarks in PyTorch Geometric, with the accompanying code available in a companion repository.

Chapter ?? examines the limitations of GNNs under the message-passing framework, and Chapter ?? synthesizes the work and outlines directions for future research.

2 Learning Tasks on Graphs

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Once a domain has been represented as graph-structured data, a natural question arises: what machine learning tasks can be defined on a graph? A graph can be viewed at three levels of granularity: its individual nodes, its edges, and the graph as a whole. Each level gives rise to a family of prediction tasks: predicting a property of each node, predicting the existence or type of a connection between pairs of nodes, or predicting a property of the entire graph. These are known, respectively, as *node-level*, *edge-level*, and *graph-level tasks* (Figure 2.1), and they structure the architectures and implementations developed in the rest of this work.

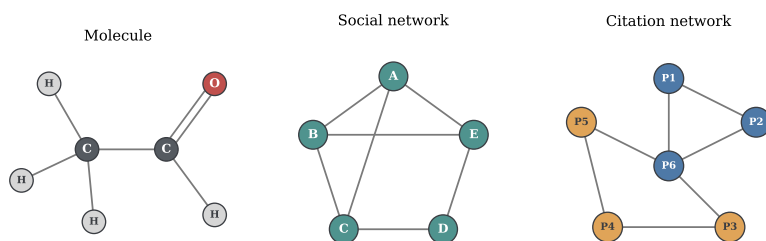
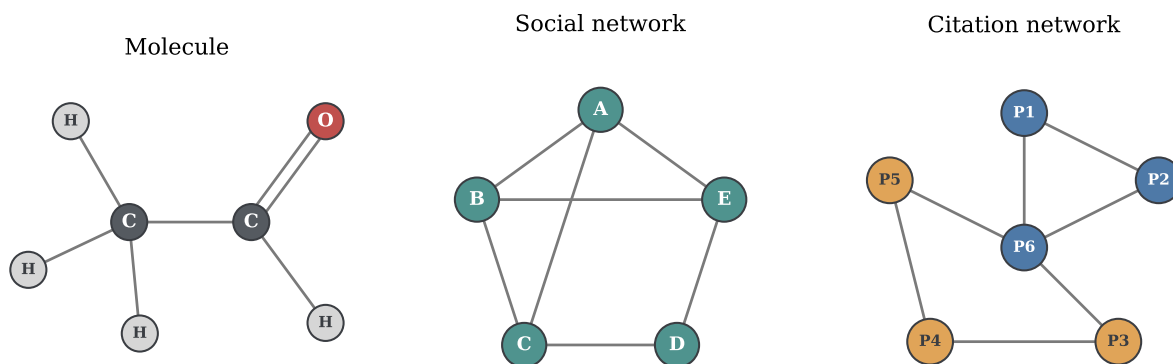


Figure 2.1: The three levels at which a prediction can be made on a single graph. A node-level task predicts a property of an individual node (highlighted); an edge-level task predicts a property of a connection between two nodes, such as whether it exists (dashed); and a graph-level task predicts a single property of the entire graph (framed). The same object supports all three.

These three levels correspond to the standard categories of graph learning tasks, but they do not exhaust the field. Other settings include *community/subgraph-level tasks*, which focus on groups of related nodes, and *graph generative tasks*, which aim to generate new graphs, such as candidate molecular structures with desirable properties. Although these settings rely on the same representational principles introduced here, they lie outside the scope of this work, which focuses on the three standard task levels.



The following sections examine each of these levels in turn and present representative applications that motivate them.

2.1 Node-Level Tasks

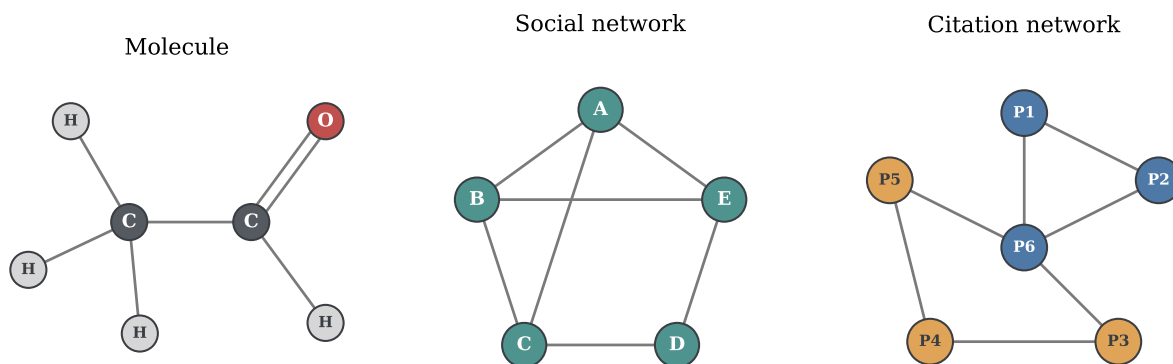
A *node-level task* asks for a prediction about each individual node of a graph, drawing on both the node's own attributes and its position within the surrounding structure. The most common instance is *node classification*: assigning each node to one of a set of categories. A representative setting is a citation network, in which the nodes are scientific papers, the edges are citations between them, and the goal is to infer the topic of each paper (Figure 2.2). A paper's own text is informative, but so is its neighborhood: papers tend to cite others on related subjects, so the topics of a paper's neighbors are strong evidence for its own. Node classification exploits exactly this, combining what a node contains with what surrounds it. The same pattern appears whenever entities in a network must be categorized.

The target need not be a category. A graph representation of a protein, for example, can be constructed where nodes represent amino acid residues and edges connect residues that are close in space. One may predict a discrete property of each residue, such as its functional role, or a continuous property, such as its position in three dimensions, a formulation that underlies recent advances in protein structure prediction¹. A node-level task can therefore be classification or regression, depending on whether the per-node target is categorical or numerical.

The same framing extends to economic settings. In a financial network whose nodes are accounts and whose edges are transactions between them, fraud detection amounts to flagging each account as legitimate or fraudulent: a node classification problem in which

Figure 2.2: Left: the node classification task in the abstract. Some nodes carry a known label (shown by color) while others are unknown (marked with a question mark); the model predicts a label for every node from its attributes and the surrounding structure. Right: a concrete instance as a citation network, where each node is a scientific paper and the predicted label is its topic.

¹ John Jumper, Richard Evans, Alexander Pritzel, Tim Green, Michael Figurnov, Olaf Ronneberger, Kathryn Tunyasuvunakool, Russ Bates, Augustin Židek, Anna Potapenko, Alex Bridgland, Clemens Meyer, Simon A. A. Kohl, Andrew J. Ballard, Andrew Cowie, Bernardino Romera-Paredes, Stanislav Nikolov, Rishub Jain, Jonas Adler, Trevor Back, Stig Petersen, David Reiman, Ellen Clancy, Michal Zielinski, Martin Steinegger, Michalina Pacholska, Tamas Berghammer, Sebastian Bodenstein, David Silver, Oriol Vinyals, Andrew W. Senior, Koray Kavukcuoglu, Pushmeet Kohli, and Demis Hassabis. Highly accurate protein structure prediction with AlphaFold. *Nature*, 596(7873): 583–589, 2021. DOI: 10.1038/s41586-021-03819-2



fraudulent behavior is often more evident in the surrounding network of connections than in any single account considered in isolation.

2.2 Edge-Level Tasks

An *edge-level task* focuses on the relationships between entities rather than the entities themselves. The central example is *link prediction*: given the edges already present in a graph, predict which further edges are likely to exist. Recommendation is one of the most familiar application (Figure 2.3). In its simplest form, a network records products that are frequently bought together, with the nodes being products and edges indicating co-purchases. Predicting missing edges in this network amounts to identifying product pairs that are likely to be associated, even if they have not yet been observed together. A richer version introduces two kinds of entity, users and products, with an edge representing interactions such as purchases or ratings; here, link prediction becomes the recommendation problem itself, anticipating which products a user will want next.

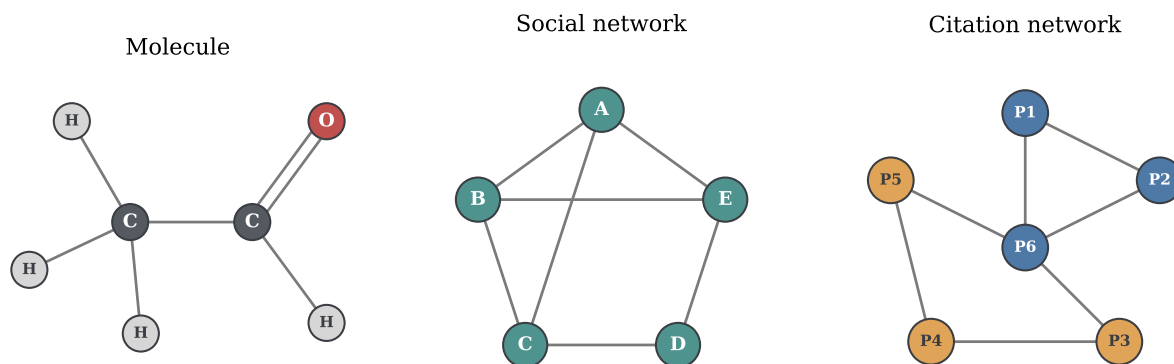
What makes the graph formulation effective is that it learns from the relational structure rather than treating each entity in isolation. A recommender system that treats products only by their individual attributes, ignoring the network of interactions around them, discards precisely the signal that reveals which items behave similarly; methods that propagate information along the graph structure consistently outperform such approaches, a principle that underlies graph-based recommender systems deployed at web scale ².

Link prediction extends well beyond commerce. Given a pair of drugs, one may predict whether taking them together produces an adverse interaction, by placing both within a larger biological network and asking whether a particular kind of connection should join them ³.

Figure 2.3: Left: the link prediction task in the abstract. Solid edges are observed connections; dashed edges marked with a question mark are candidate connections whose existence the model must predict. Right: a concrete instance as a recommendation setting, where the two kinds of entity (users, drawn as squares, and products, drawn as circles) are joined by observed interactions, and the task is to predict further user-product links.

² Rex Ying, Ruining He, Kaifeng Chen, Pong Eksombatchai, William L. Hamilton, and Jure Leskovec. Graph convolutional neural networks for web-scale recommender systems. In *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining (KDD)*, pages 974–983, 2018. DOI: 10.1145/3219819.3219890

³ Marinka Zitnik, Monica Agrawal, and Jure Leskovec. Modeling polypharmacy side effects with graph convolutional networks. *Bioinformatics*, 34(13):i457–i466, 2018. DOI: 10.1093/bioinformatics/bty294



The same formulation is also used to complete *knowledge graphs*, large networks of facts in which entities are connected by relations: many true relations are simply unobserved, and link prediction proposes the most plausible ones.

2.3 Graph-Level Tasks

A *graph-level task* treats an entire graph as a single example and predicts a property of the whole. The representative task is *graph classification*: assigning a label to a complete graph (Figure 2.4). Here the data is not one large network but a collection of many separate graphs, each with its own label, and the model must summarize the structure and attributes of an entire graph into a single prediction. Molecular property prediction is the canonical example. A molecule is naturally a graph, with atoms as nodes and chemical bonds as edges, and many prediction tasks about it are graph-level: whether a compound is toxic, or whether it is active against a given biological target. Because molecules vary in size and structure, this is exactly the regime that calls for a model able to handle graphs of varying structure and return one prediction per graph.

This capability feeds directly into drug discovery, where large libraries of candidate compounds are screened to prioritize the few worth synthesizing and testing in the laboratory, a process in which an accurate graph-level predictor can save considerable experimental effort. The same idea applies in materials science, where the atomic structure of a material is represented as a graph and a graph-level model predicts a physical property such as its stability or its mechanical strength.

Across all three levels, the graph is used in different ways, with the level of the task determining what is predicted and how a model must process the structure to predict it.

Figure 2.4: Left: the graph classification task in the abstract. The data is a collection of distinct graphs, differing in size and shape, each assigned a single label for the graph as a whole. Right: a concrete instance as molecular property prediction, where each graph is a molecule (atoms as nodes, bonds as edges) labeled by a property such as being active or inactive against a target.

3 Fundamentals of Graph Theory

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The applications surveyed in Chapter 2 share a single underlying object: the graph. To move from describing tasks to formulating them precisely, that object must itself be defined formally. This chapter introduces the formal language of graphs used throughout the rest of this work. It first defines graphs, neighborhoods, and node and edge features; it then encodes graph structure algebraically through the adjacency matrix and distinguishes the main types of graphs; it examines the structural properties, from walks to isomorphism, on which later developments rely; it introduces the graph Laplacian and the spectral perspective it provides; and it closes with a summary of the notation fixed here.

3.1 Formal Definitions

A *graph* is a pair

$$\mathcal{G} = (V, E), \tag{3.1}$$

where V is a finite set of *nodes*, also called vertices, and E is a set of *edges*, also called links, each one an unordered pair of nodes recording which two nodes are connected ¹. The number of nodes is denoted $n = |V|$ and the number of edges $m = |E|$. In classical graph-theoretic notation, generic nodes are denoted by $u, v \in V$, and an edge between them by the unordered pair $\{u, v\} \in E$; at this stage, the notation simply states that the two nodes are connected.

Because V and E are sets in the mathematical sense, neither carries an intrinsic order: a graph specifies which nodes are connected, and

¹ Reinhard Diestel. *Graph Theory*, volume 173 of *Graduate Texts in Mathematics*. Springer, 5 edition, 2017

nothing more. It is nevertheless convenient, particularly when graphs are represented by matrices, to fix an indexing of the nodes by integers. Throughout this work, nodes are therefore denoted $i, j \in \{1, \dots, n\}$, with the convention that i refers to the node under consideration and j to a neighboring node whenever a neighborhood relation is involved. The choice of indexing is arbitrary and carries no information about the graph itself, a point this chapter returns to.

Two nodes are *adjacent*, or *neighbors*, when an edge connects them. The *neighborhood* of a node i is the set of its neighbors,

$$\mathcal{N}(i) = \{j \in V : \{i, j\} \in E\}, \quad (3.2)$$

and the *degree* of node i is the number of neighbors it has,

$$\deg(i) = |\mathcal{N}(i)|. \quad (3.3)$$

In a social network, for example, the neighborhood of a person is the set of people connected to them, and the degree counts them.

Beyond its structure, a graph typically carries data ². Each node i has an associated *node feature vector* $\mathbf{x}_i \in \mathbb{R}^d$, where d is the node feature dimension, describing the attributes of the entity the node represents, such as the textual content of a paper in a citation network, or the chemical element of an atom in a molecule. Stacking the n node feature vectors as rows produces the *node feature matrix*

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_1^\top \\ \vdots \\ \mathbf{x}_n^\top \end{bmatrix} \in \mathbb{R}^{n \times d}, \quad (3.4)$$

whose i -th row is the transposed feature vector of node i . Writing the features as a matrix presupposes that the node indexing is fixed as above. Edges can carry attributes as well: the *edge feature vector* $\mathbf{e}_{i,j} \in \mathbb{R}^c$, where c is the edge feature dimension, describes the connection between nodes i and j , such as the type of a chemical bond in a molecule.

² William L. Hamilton. *Graph Representation Learning*. Synthesis Lectures on Artificial Intelligence and Machine Learning. Morgan & Claypool, 2020

3.2 The Adjacency Matrix and Graph Types

With the nodes indexed from 1 to n , the structure of a graph can be encoded in a single matrix. The *adjacency matrix* of $\mathcal{G}$ is the matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ with entries

$$A_{ij} = \begin{cases} 1 & \text{if } \{i, j\} \in E, \\ 0 & \text{otherwise,} \end{cases} \quad (3.5)$$

so that each entry records the presence or absence of a single edge.

The adjacency matrix depends on the node indexing. Reordering

There are $n!$ possible indexings of n nodes, and each yields a valid adjacency matrix of the same graph.

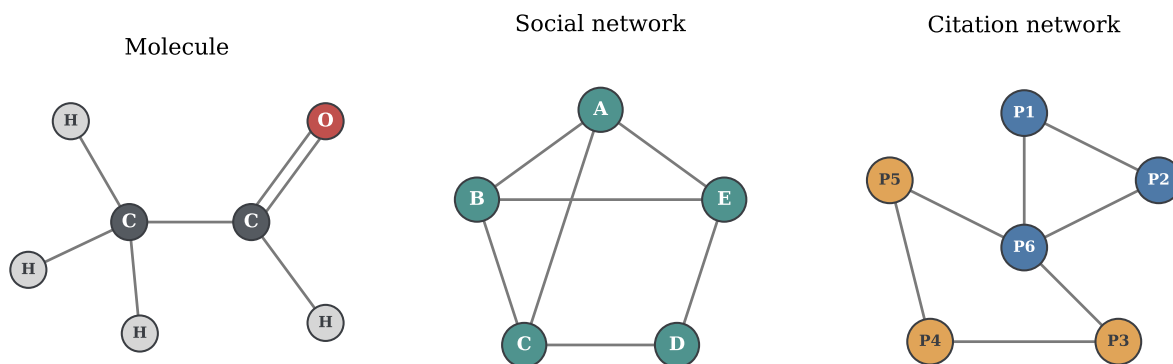


Figure 3.1: A graph on six nodes (left) and its adjacency matrices under two different indexings of the nodes (center and right). Both matrices describe the same structure: reordering the nodes permutes the rows and columns of $\mathbf{A}$.

the nodes permutes the rows and columns of $\mathbf{A}$, producing a different matrix that describes exactly the same graph. Figure 3.1 shows a graph together with its adjacency matrices under two different indexings: a graph has not one adjacency matrix but one per indexing, and no single one of them is privileged.

The first distinction among graph types concerns the orientation of edges. In an *undirected* graph, the setting assumed so far, an edge $\{i, j\}$ has no orientation and can be traversed in either direction; its adjacency matrix is consequently symmetric, $\mathbf{A} = \mathbf{A}^\top$. In a *directed* graph, edges are instead ordered pairs: the pair $(i, j) \in E$ denotes an edge from a *source* node i to a *target* node j , and the presence of (i, j) no longer implies the presence of (j, i) . The adjacency entry reads $A_{ij} = 1$ exactly when $(i, j) \in E$, so $\mathbf{A}$ is in general not symmetric and row i lists the targets of the edges leaving node i . The degree of a node accordingly splits in two: its *out-degree*, the number of edges leaving it, and its *in-degree*, the number of edges arriving at it. Unless stated otherwise, the graphs treated in this work are undirected.

A second distinction concerns whether edges carry a magnitude. In a *weighted* graph, each edge has an associated scalar weight $w_{ij} \in \mathbb{R}$, expressing, for instance, the strength of a relationship or a physical distance; the adjacency matrix then stores $A_{ij} = w_{ij}$ in place of 1 (Figure 3.2). In an *unweighted* graph, every edge counts the same and (3.5) applies as stated. A scalar weight can be viewed as the special case $c = 1$ of the edge feature vectors introduced in Section 3.1, that is, a single number attached to each edge.

A *self-loop* is an edge connecting a node to itself; in the adjacency matrix it appears as a nonzero diagonal entry, $A_{ii} \neq 0$. A *multigraph* allows several parallel edges between the same pair of nodes, a multiplicity that the binary entries of (3.5) cannot record. A graph with no self-loops and at most one edge between any pair of nodes

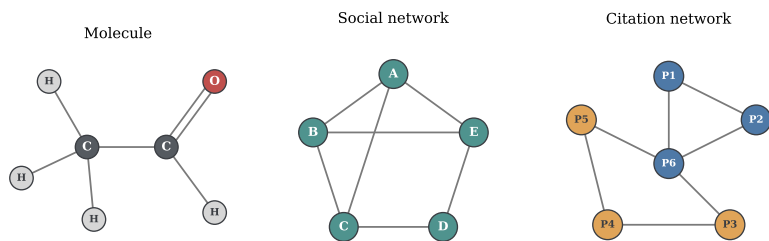


Figure 3.2: Left: a directed version of the graph of Figure 3.1 and its adjacency matrix. Edge orientations make $\mathbf{A}$ asymmetric: row i lists the targets of the edges leaving node i , and the in-degree and out-degree of one highlighted node are annotated. Right: a weighted graph, with edge thickness proportional to the weight w_{ij} .

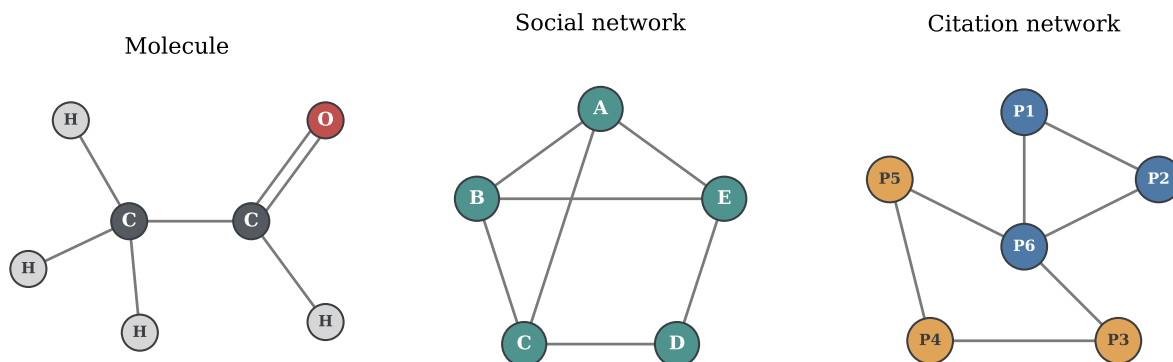


Figure 3.3: Left: a graph with self-loops. Center: a multigraph with parallel edges. Right: a bipartite graph, with its two subsets of nodes drawn as distinct shapes.

is called *simple*. A *bipartite* graph, in contrast, is not a departure from simplicity but a structural subclass of it: its nodes partition into two disjoint subsets, and every edge connects a node of one subset with a node of the other, as in a network of users and products joined by purchases. Because of this specificity, such graphs are explicitly named bipartite; a graph described only as simple implies no such partition. Figure 3.3 illustrates the three cases. As established in Chapter 1, the graphs treated in this work are simple.

The last distinction concerns the kinds of entities and relations a graph contains. A graph is *homogeneous* when all of its nodes represent one kind of entity and all of its edges represent one kind of relation: every node then shares the same feature space $\mathbb{R}^d$ and every edge the same feature space $\mathbb{R}^c$. A citation network is homogeneous, since every node is a paper and every edge a citation. A graph is *heterogeneous* when it contains several types of nodes or edges, each with its own semantics and, typically, its own feature space³. A citation network that includes both papers and their authors is heterogeneous: papers and authors are different kinds of entities, joined by a relation of authorship. Figure 3.4 contrasts the two settings. As stated in Chapter 1, this work treats homogeneous graphs.

The homogeneous/heterogeneous terminology comes from the network analysis and machine learning literature rather than from classical graph theory.

³ William L. Hamilton. *Graph Representation Learning*. Synthesis Lectures on Artificial Intelligence and Machine Learning. Morgan & Claypool, 2020

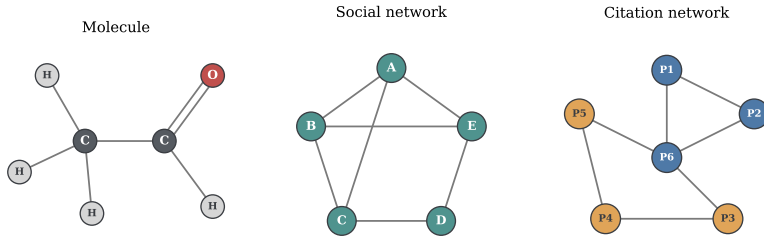


Figure 3.4: Left: a homogeneous graph, in which all nodes represent the same kind of entity and all edges the same relation. Right: a heterogeneous graph, in which node shapes denote different entity types and edge styles denote different relations.

3.3 Walks, Connectivity, Homophily, and Isomorphism

The definitions so far describe a graph locally: nodes, their neighbors, and their attributes. Several properties that matter in the remainder of this work are instead global, arising from how edges chain together across the graph. A *walk* is a sequence of nodes in which each consecutive pair is joined by an edge, and its *length* is the number of edges traversed; a walk may revisit nodes and edges. A *path* is a walk in which all nodes are distinct (Figure 3.5).

The *geodesic distance* between two nodes is the length of a shortest path between them,

$$d(i, j) = \min \left\{ k \in \mathbb{N} : \text{there is a path of length } k \text{ between } i \text{ and } j \right\} \quad (3.6)$$

with $d(i, i) = 0$. The distance is always written with its two arguments, $d(i, j)$, which distinguishes it from the node feature dimension d . When no path joins i and j , the distance is taken to be infinite, a situation characterized next.

A graph is *connected* when a path exists between every pair of its nodes, that is, when $d(i, j)$ is finite for all $i, j \in V$. When it is not, the graph decomposes into *connected components*: maximal groups of nodes that are mutually reachable, with no edges between groups (Figure 3.6). In directed graphs, connectivity refines into weak and strong variants according to whether edge orientations are respected; since this work treats undirected graphs, the refinement is not developed further.

Structure and attributes are often correlated. A graph exhibits *homophily* when adjacent nodes tend to have similar attributes, and *heterophily* when they tend to differ. The notion originates in the analysis of social networks, where similar individuals form ties at higher rates than dissimilar ones ⁴, and it extends to many of the graphs of Chapter 2: in a citation network, a paper predominantly cites papers on related topics, so neighboring nodes tend to share a subject. Whether a graph is homophilous or heterophilous is an

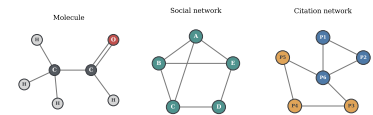


Figure 3.5: A walk that revisits a node (top) and a shortest path between the same endpoints (bottom), whose length is their geodesic distance.

⁴ Miller McPherson, Lynn Smith-Lovin, and James M. Cook. Birds of a feather: Homophily in social networks. *Annual Review of Sociology*, 27:415–444, 2001. DOI: 10.1146/annurev.soc.27.1.415

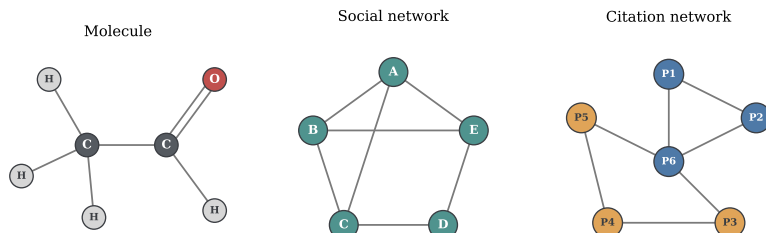


Figure 3.6: A disconnected graph with two connected components, one of which is a single isolated node.

empirical property of the data rather than a structural constraint, and the degree to which it holds varies from one graph to another.

The arbitrariness of node indexing observed in Section 3.2 raises a final question: when do two graphs that are presented differently have the same structure? Two graphs $\mathcal{G}_1 = (V_1, E_1)$ and $\mathcal{G}_2 = (V_2, E_2)$ are *isomorphic* when there exists a bijection $\pi : V_1 \rightarrow V_2$ that preserves adjacency,

$$\{i, j\} \in E_1 \iff \{\pi(i), \pi(j)\} \in E_2, \quad (3.7)$$

for all nodes $i, j \in V_1$. Isomorphic graphs are the same graph up to a relabeling of its nodes; the two adjacency matrices of Figure 3.1 are representations related in exactly this way. Two graphs drawn very differently may be isomorphic (Figure 3.7). Determining whether two graphs are isomorphic is a fundamental problem in graph theory. A classical heuristic, the Weisfeiler–Leman test, distinguishes most non-isomorphic graphs in practice ⁵, and it reappears later in this work in connection with the expressive power of graph neural networks.

3.4 The Graph Laplacian and the Spectral Perspective

The structural properties of Section 3.3 and the algebraic encoding of Section 3.2 meet in a single observation: powers of the adjacency matrix count walks. For any positive integer k , the entry in row i and column j of the k -th matrix power gives the number of walks of length k between the two nodes,

$$(\mathbf{A}^k)_{ij} = |\{\text{walks of length } k \text{ between } i \text{ and } j\}|, \quad (3.8)$$

a count that is symmetric in i and j for the undirected graphs treated here. In particular, the entry is nonzero exactly when the two nodes are joined by some walk of that length, which ties the reachability notions of Section 3.3 directly to matrix algebra and motivates the matrices introduced next.

The *degree matrix* $\mathbf{D} \in \mathbb{R}^{n \times n}$ is the diagonal matrix that holds the node degrees,

$$\mathbf{D} = \text{diag}(\deg(1), \dots, \deg(n)), \quad (3.9)$$

⁵ Christopher Morris, Yaron Lipman, Haggai Maron, Bastian Rieck, Nils M. Kriege, Martin Grohe, Matthias Fey, and Karsten Borgwardt. Weisfeiler and Leman go machine learning: The story so far. *Journal of Machine Learning Research*, 24(333):1–59, 2023

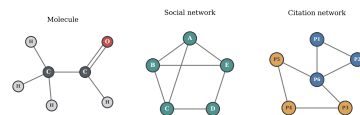


Figure 3.7: Two graphs that are drawn differently but are isomorphic: a bijection between their node sets preserves all adjacencies.

with every off-diagonal entry equal to zero. Combining it with the adjacency matrix yields the central operator of spectral graph theory ⁶, the *graph Laplacian*

$$\mathbf{L} = \mathbf{D} - \mathbf{A}. \quad (3.10)$$

Each diagonal entry of $\mathbf{L}$ is the degree of the corresponding node, and each off-diagonal entry is -1 where an edge is present and 0 otherwise. For the six-node graph of Figure 3.1, under the node indexing of its centre panel and with $\mathbf{D} = \text{diag}(2, 3, 2, 3, 3, 1)$, the Laplacian is

$$\mathbf{L} = \begin{bmatrix} 2 & -1 & 0 & 0 & -1 & 0 \\ -1 & 3 & -1 & 0 & -1 & 0 \\ 0 & -1 & 2 & -1 & 0 & 0 \\ 0 & 0 & -1 & 3 & -1 & -1 \\ -1 & -1 & 0 & -1 & 3 & 0 \\ 0 & 0 & 0 & -1 & 0 & 1 \end{bmatrix}.$$

The Laplacian admits a direct interpretation in terms of how values spread over a graph. Assigning a real number to each node defines a vector $\mathbf{f} \in \mathbb{R}^n$ called a *graph signal*, with component f_i the value at node i . The Laplacian measures how much such an assignment varies across edges through the quadratic form

$$\mathbf{f}^\top \mathbf{L} \mathbf{f} = \sum_{\{i,j\} \in E} (f_i - f_j)^2, \quad (3.11)$$

which sums the squared difference between the values at the two endpoints of every edge. The form vanishes when $\mathbf{f}$ is constant on each connected component and grows as the values assigned to neighboring nodes diverge. Because it is a sum of squares, the quadratic form is nonnegative for every $\mathbf{f}$, so the Laplacian is *positive semidefinite*.

Being real and symmetric, the Laplacian admits an eigendecomposition

$$\mathbf{L} = \mathbf{U} \mathbf{\Lambda} \mathbf{U}^\top, \quad (3.12)$$

where the columns of $\mathbf{U} \in \mathbb{R}^{n \times n}$ form an orthonormal basis of eigenvectors and $\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_n)$ collects the corresponding eigenvalues. Symmetry guarantees that these eigenvalues are real, and positive semidefiniteness that they are nonnegative; ordered, they satisfy $0 = \lambda_1 \leq \dots \leq \lambda_n$. The smallest eigenvalue is always zero, with the constant vector as an eigenvector, since the rows of $\mathbf{L}$ sum to zero. More can be said: the multiplicity of the eigenvalue 0 equals the number of connected components of the graph, linking the spectral description back to the connectivity of Section 3.3.

For many purposes the Laplacian is rescaled by the node degrees. The *symmetric normalized Laplacian* is

$$\mathbf{L}_{\text{sym}} = \mathbf{D}^{-1/2} \mathbf{L} \mathbf{D}^{-1/2} = \mathbf{I} - \mathbf{D}^{-1/2} \mathbf{A} \mathbf{D}^{-1/2}, \quad (3.13)$$

⁶ Fan R. K. Chung. *Spectral Graph Theory*, volume 92 of *CBMS Regional Conference Series in Mathematics*. American Mathematical Society, 1997

where $\mathbf{I}$ is the identity matrix and $\mathbf{D}^{-1/2}$ is the diagonal matrix of inverse square-root degrees. It is again symmetric and positive semidefinite, so it shares the spectral structure just described, while its eigenvalues are confined to the interval $[0, 2]$ ⁷. This normalized form is the variant used in the spectral developments of later chapters. Together, these properties make the Laplacian a natural bridge between the structure of a graph and the linear algebra used to operate on it.

For a node of degree zero the corresponding entry of $\mathbf{D}^{-1/2}$ is taken to be zero by convention.

⁷ Fan R. K. Chung. *Spectral Graph Theory*, volume 92 of *CBMS Regional Conference Series in Mathematics*. American Mathematical Society, 1997

3.5 Notation Summary

The notation introduced in this chapter is collected in Table 3.1 for reference. It is used consistently throughout the remainder of this work.

Symbol	Meaning	Reference
$\mathcal{G} = (V, E)$	Graph with node set V and edge set E	(3.1)
n, m	Number of nodes and edges	§3.1
i, j	Nodes; i central, j a neighbor	§3.1
$\mathcal{N}(i)$	Neighborhood of node i	(3.2)
$\deg(i)$	Degree of node i	(3.3)
$d(i, j)$	Geodesic distance between i and j	(3.6)
π	Isomorphism bijection between node sets	(3.7)
$\mathbf{x}_i, d$	Node feature vector and its dimension	(3.4)
$\mathbf{X}$	Node feature matrix	(3.4)
$\mathbf{e}_{i,j}, c$	Edge feature vector and its dimension	§3.1
w_{ij}	Scalar edge weight	§3.2
$\mathbf{A}$	Adjacency matrix	(3.5)
$(\mathbf{A}^k)_{ij}$	Walks of length k between i and j	(3.8)
$\mathbf{D}$	Degree matrix	(3.9)
$\mathbf{L}$	Graph Laplacian	(3.10)
$\mathbf{L}_{\text{sym}}$	Symmetric normalized Laplacian	(3.13)
$\mathbf{f}$	Graph signal, one real value per node	(3.11)
$\mathbf{U}, \mathbf{\Lambda}$	Eigenvectors and eigenvalues of $\mathbf{L}$	(3.12)
$\lambda_1, \dots, \lambda_n$	Laplacian eigenvalues, in increasing order	(3.12)

Table 3.1: Notation introduced in this chapter, with the equation or section in which each symbol is defined.

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